

Practical Demonstration of MERMAID Framework and Low Compute Extension

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I. PROJECT DETAILS

This project follows the criterion of reproducing an existing research system and then addressing a clearly identified limitation. We will reproduce the core inference-time pipeline of MERMAID (Multi-perspective Emotion Recognition via Multi-Agent generative Decision-making) [1] and extend it in two directions: (i) demographic and cultural robustness, motivated by the Western-centric and demographically limited benchmarks identified by the paper, and (ii) compute-efficient execution on a commodity system with 32 GB RAM and an NVIDIA RTX 3050 with 12 GB VRAM. The original work reports improvements of up to 27.90 percentage points over strong baselines and explicitly identifies demographic/cultural fairness as underexplored.

Team Composition and Individual Contributions

- **Pulak Kumar Sarkar (B24DS505)**

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Responsible for model loading and configuration, integrating the MLLM used for emotion classification, implementing the decision-agent inference pipeline, and integrating the Q-Former with the overall framework. He will also support baseline reproduction and evaluation.

- **Umesh Vishwakarma (B24CS507)**

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Textual Self-Reflection Agent: Responsible for agent initialization, prompt design and tuning, implementation of the textual self-reflection agent, feedback generation, and state transitions between reflection and decision stages. He will also investigate LCM-based acceleration of the generative component.

- **Madan Vishwakarma (B24DS504)**

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Visual Self-Reflection Agent: Responsible for implementing the visual self-reflection module, comparing the original image with emotion-conditioned reference images, generating visual feedback, and integrating the visual-reflection cycle with the decision agent. He will also handle systems-level optimization and end-to-end deployment on the target hardware.

II. PROBLEM STATEMENT

Multimodal large language models (MLLMs) show strong visual-language reasoning ability but remain difficult to trust for subtle emotion recognition. MERMAID addresses this problem by combining a decision agent, textual and visual self-reflection agents, and emotion-guided visual augmentation in an iterative cross-modal workflow. However, the paper explicitly states that its fairness under demographic or cultural

variation remains underexplored, while benchmarks such as EmoSet and Emotion6 are described as Western-centric or demographically limited. The authors further note that pretrained generative models may reproduce stereotypical associations between affective states and demographic traits.

Our objective is therefore to develop a practical extension of MERMAID that retains the original framework’s reasoning benefits while testing and reducing demographic and cultural disparities. We will additionally reproduce the method as closely as possible and target performance close to the reported MERMAID accuracy before evaluating the fairness extension. A second practical objective is to make the pipeline executable on a substantially smaller compute budget than the paper’s reported H100-based setup.

III. METHODOLOGY

A. Baseline Reproduction

The initial implementation will reproduce the MERMAID inference pipeline: an MLLM decision agent produces an initial emotion prediction; a textual self-reflection agent evaluates the prediction from multiple perspectives; an emotion-guided augmentation agent generates candidate visual exemplars; and a visual self-reflection agent compares the original image with the generated emotion-conditioned references. Feedback is routed back to the decision agent until an early-exit criterion is satisfied or a maximum number of iterations is reached. The paper uses frozen instruction-tuned MLLMs and reports a maximum of three outer iterations.

B. Compute-Efficient MERMAID

The original system uses 2B–13B MLLMs and generates emotion-conditioned images with InstructPix2Pix [2] using 100 denoising steps, with an average end-to-end processing time of approximately 15 seconds per image on an NVIDIA H100. Our implementation will preserve the logical pipeline while reducing the memory footprint.

First, we will prioritise Qwen2-VL-2B for the fully local baseline and investigate 4-bit/8-bit quantisation for a 7B model when memory permits. Second, the MLLM and diffusion components will not remain resident on the GPU simultaneously. We will use sequential loading/offloading, mixed precision, cached intermediate outputs, explicit CUDA memory cleanup, and small batch sizes. Third, the image-generation stage will be investigated with few-step acceleration using a latent-consistency approach such as LCM/LCM-LoRA [3] or another compatible distilled Stable Diffusion [4] scheduler. The target is to reduce generation from roughly 100 denoising steps to

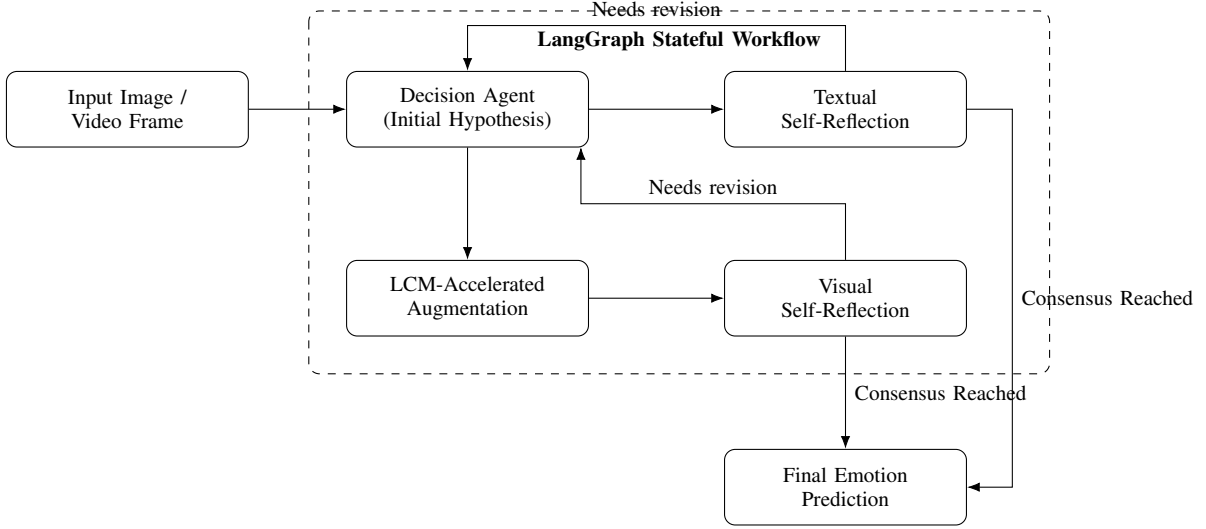


Fig. 1. Planned coordinated workflow for the compute-efficient MERMAID extension. The architecture leverages LCM-accelerated emotion-guided augmentation for rapid inference on constrained hardware.

approximately 2–4 steps while measuring whether the generated exemplars remain sufficiently faithful for visual reflection. This acceleration is treated as an experimental substitution rather than an assumption of equal accuracy.

C. LangGraph-Based Orchestration

The multi-agent interaction will be implemented as a stateful graph instead of a large imperative loop. Nodes will represent the decision, textual reflection, demographic/cultural moderation, augmentation, and visual reflection agents. Conditional edges will implement early exits and revision cycles. Shared graph state will store the image identifier, candidate emotions, current prediction, textual feedback, visual feedback, demographic fairness feedback, iteration count, and profiling statistics. This design is intended to make the cyclic workflow easier to test, debug, reproduce, and modify.

IV. RESOURCE AND DATASET AVAILABILITY

A. Datasets

EmoSet. We will retain EmoSet [5] as the principal reproduction benchmark because it is the main large-scale dataset used in MERMAID. EmoSet contains 3.3 million images, including 118,102 human-labeled images, and provides emotion labels together with rich visual attributes. It will be used to establish the reproduction baseline and to compare accuracy against the reported results.

Emotion6 and Artphoto. These datasets will be retained for out-of-domain and stylistically diverse evaluation, following the original MERMAID experimental design. They will help test whether any fairness-oriented modification harms generalisation.

LAI-GAI [6] is a geographically and culturally diverse affective image dataset designed to study variations in human emotion perception. It contains **847** affective images representing **12** affective states, with evaluations collected from **more than 2,400** participants across **58** countries, grouped into six

cultural cohorts. Unlike face-centric emotion datasets, LAI-GAI uses diverse visual stimuli and provides cross-cultural affective ratings, making it suitable for evaluating whether emotion recognition models generalize fairly across different geographic and cultural populations.

B. Preprocessing

Images will be resized to the model input resolution and normalised using the selected model processor. Face-centric samples will be retained with sufficient context so that the visual self-reflection module can use both expression and scene information. Emotion labels will be mapped only when a defensible semantic correspondence exists; otherwise, the original annotation space will be reported separately. Subgroup metadata will never be inferred as ground truth when reliable dataset annotations are unavailable.

C. Hardware and Software

- The original MERMAID framework relies on enterprise-grade hardware, such as the NVIDIA H100, to execute its heavy multi-agent pipeline in roughly 15 seconds per image. Deploying this on your highly constrained local system (Intel i7, 32GB RAM, and an RTX 3050 with 12GB VRAM) requires aggressive memory management like 4-bit quantization and CPU/GPU offloading. While these workarounds prevent memory crashes, quantization will slightly degrade the model’s raw reasoning accuracy, and running the generative agent in fewer steps via FP16 will introduce minor visual artifacts compared to the pristine outputs of an H100.
- Here are the major libraries:
 - `torch & torchvision`: Core machine learning framework and CUDA memory clearing.
 - `accelerate`: Managing model offloading between your CPU RAM and GPU VRAM.

- `bitsandbytes`: Quantizing the Vision-Language Model to 4-bit so it fits on your RTX 3050.
- `transformers`: Loading and running the decision and reflection models.
- `diffusers` & `peft`: Running the image generation and injecting the LCM-LoRA.
- `langgraph` & `langchain-core`: Building the cyclic graph and managing the state payload between agents.
- `instructor` (or `outlines`): Forcing the models to output strictly formatted JSON and the discrete +1/0 evaluation scores.

V. LIMITATIONS ADDRESSED

The central limitation addressed from MERMAID is its underexplored fairness under demographic and cultural variation. The paper states that EmoSet and Emotion6 are Western-centric or demographically limited and warns that pretrained generative models may reinforce stereotypical emotional associations. To address MERMAID’s Western-centric evaluation limitation, we will incorporate LAI-GAI, which contains 847 affective images representing 12 affective states and cross-cultural evaluations from more than 2,400 participants across 58 countries grouped into six cultural cohorts.

A second limitation of MERMAID is its high computational cost due to repeated MLLM calls and emotion-conditioned image generation using 100 diffusion denoising steps. This makes the framework difficult to reproduce on low-resource hardware such as our RTX 3050 (12 GB VRAM) with 32 GB RAM. To address this, we will investigate Latent Consistency Model (LCM)-based acceleration for the InstructPix2Pix augmentation stage, reducing the number of denoising steps substantially. This should lower inference time, memory usage, and overall computational overhead while retaining the role of visual augmentation in MERMAID.

However, aggressive acceleration may affect the quality or emotional fidelity of generated reference images, which could in turn influence visual self-reflection and final predictions. Therefore, the accelerated implementation will be evaluated against the original approach using recognition accuracy, inference time, memory consumption, and visual consistency. Our objective is to identify a practical configuration that enables MERMAID to run efficiently on limited hardware while keeping its recognition performance as close as possible to the accuracy reported in the original paper.

VI. EXPECTED OUTCOMES

The project is expected to deliver the following outcomes:

- A reproducible local implementation of the MERMAID inference pipeline, with accuracy targeted as close as practical to the original paper’s reported results under comparable settings.
- A demographic / cultural moderation extension with subgroup-aware evaluation and fairness reporting.
- A low-resource implementation capable of running within approximately 12 GB VRAM and 32 GB RAM through

quantisation, sequential offloading, caching, and memory-aware orchestration.

- An experimental few-step generative augmentation variant targeting approximately 2–4 denoising steps, together with an accuracy/latency/VRAM trade-off analysis.
- A LangGraph implementation of the cyclic multi-agent workflow, enabling explicit state tracking, conditional routing, iteration limits, and reproducible experiments.
- Final ablation studies comparing the original reproduction, the fairness extension, the compute-efficient version, and the combined system.

The primary success criterion is not simply a higher aggregate accuracy. We aim to obtain a three-way trade-off: competitive emotion-recognition accuracy, reduced inference cost, and smaller performance disparities across demographic/cultural groups. The final report will document where this trade-off succeeds and where the proposed modifications introduce measurable degradation.

REFERENCES

- [1] Z. Yang, J. Song, S. Song, W. Pang, and Y. Yuan, “Mermaid: Multi-perspective self-reflective agents with generative augmentation for emotion recognition,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 24 639–24 655.
- [2] T. Brooks, A. Holynski, and A. A. Efros, “Instructpix2pix: Learning to follow image editing instructions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 18 392–18 402.
- [3] S. Luo, Y. Tan, L. Huang, J. Li, and H. Zhao, “Latent consistency models: Synthesizing high-resolution images with few-step inference,” *arXiv preprint arXiv:2310.04378*, 2023.
- [4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 10 684–10 695.
- [5] J. Yang, Q. Huang, T. Ding, D. Lischinski, D. Cohen-Or, and H. Huang, “Emoset: A large-scale visual emotion dataset with rich attributes,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 20 383–20 394.
- [6] M. Behnke, M. Kłoskowski, M. Klichowski, W. Krzyżaniak, K. Szymański, P. Maciejewski, P. Chwiłkowska, M. Kowal, R. Jończyk, J. Nowak, S. Kupiński, D. Kunc, S. Saganowski, A. A. Chowkase, F. Guemaz, K. S. Kertechian, A. I. M. T. Maadal, L. A. Aguilar, B. T. Alayande, V. Balakrishnan, D. M. Basnight-Brown, J. Boudesseul, T. A. D’Amelio, J. C. Dacanay, A. Dedhe, S. Gao, J. F. G. B. Takayanagi, M. R. Islam, A. Mailhos, C. M. Mpyangu, M. Mebarak, A. Najmussaib, J. H. Park, E. Pirtskhalava, E. Rice, S. Sami, Y. Yamada, J. Baczyński, L. Dera, S. Jesko-Białek, J. Łączkowski, H. Marciniak, F. Nowicki, B. Wilczek, J. J. Gross, and N. A. Coles, “Using artificial intelligence to generate affective images: Methodology and initial library,” *Advances in Methods and Practices in Psychological Science*, vol. 9, no. 1, 2026.